AL AMRU BIL MARUF

Software Engineer II

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Dhaka, Bangladesh mortytheplatypus

SKILLS

Languages:	`Java, JavaScript, Go, Python, C/C++, Bash`
Frameworks:	`Spring Boot, React, Next.js, Express.js (NodeJS), Flask`
Databases:	`MySQL, PostgreSQL, MongoDB, Cassandra`
Infrastructure:	`Nginx, Docker, Kafka, Redis, RabbitMQ, CDN, RTMP, HLS, Cloudflare R2`
Tools/Protocols:	`Git, Liquibase, Locust, WebSocket, XMPP, JWT, Eureka, Consul`

EXPERIENCE

June 2024 onwards

Synesis IT PLC — Software Engineer

Core contributor to Convay, a high-scale video conferencing platform.

- **Streaming at Scale:** Designed a gated multi-origin HLS platform (control plane, race-safe origin pool, Cloudflare R2 offload) so live delivery scales past origin limits–JWT playlist/segment auth and abuse-resistant ownership caching under CDN load; Locust-validated at 10k concurrent viewers with near-zero playlist failure on healthy paths.
- **Realtime Messaging:** Built Spring Boot chat microservices on Kafka, Cassandra, and Redis with presence batching and WebSocket fan-out for multi-thousand concurrent participants; soak-tested to 4k STOMP connections per instance.
- **Webinar Protocol:** Shipped hosts-only guest messaging end-to-end across the signaling layer, realtime client SDK, and web UI, including history replay so late joiners and promoted hosts stay in sync in large rooms.
- **Meeting Orchestration:** Developed big-meeting livestream initiation with single-flight concurrency control, streaming origin pool acquire/release and failover, and HMAC-signed enterprise guest join links.
- **Usage Billing:** Built a Cloudflare Analytics pipeline that aggregates concurrent viewers and viewer-minutes on lagged sync windows, with failure backfill and VOD delivery metrics for stream monetization.

Tech Stack: `Spring Boot, Go, React, Kafka, Cassandra, Redis, MySQL, Docker, Cloudflare R2, Locust`

May 2023 — June 2023

MYSOFT LIMITED — Health Analytics Intern

Developed a real-time health analytics dashboard to visualize hospital data trends.

- Processed multi-dimensional health datasets to identify regional and seasonal disease outbreaks based on patient demographics (age, gender, location).
- Built a full-stack application using Flask and Angular to provide hospital administrators with actionable insights into operational overhead.

Tech Stack: `Flask, Angular, PostgreSQL`

PROJECTS

`Go`

Gozzle — GitHub

A log-ingestion service with per-service WAL storage, bounded async queues with backpressure, offset-based reads, SSE live tailing, log rotation and retention.

`PostgreSQL` `Spring Boot`

Jwt Authenticator — GitHub

A reusable full-stack authentication template providing secure registration and session management using JSON Web Tokens (JWT), designed for rapid integration into new web applications.

`Next.js` `Express.js`

Find My Apartment — Backend | Frontend

An apartment-search platform with advanced filtering, personalized recommendation logic, and integrated map views for enhanced user experience.

`Flask` `React`

Pasteruf — GitHub

A dual-purpose platform for persistent text storage and URL shortening. Features automatic redirection logic and a streamlined interface for sharing content across platforms.

`PLSQL` `React` `Express.js`

Fantasy Premier League — Fantasy Premier League GitHub

Replica of Fantasy Premier League, which allows users to create fantasy football teams, select players, and compete in leagues.

EDUCATION

2019 — 2024 **Bangladesh University of Engineering and Technology** **BSc in CSE**
CGPA: 3.48/4.00
Thesis: Handwritten Bangla Character Recognition Using Convolutional Neural Networks (CNN)

OTHERS

- Fluent in Bengali and English
- Senior Member and Mentor in Dinajpur Math Club